CivicResolve AI

Municipal Operational Intelligence & Delay Risk Diagnostic

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1. Executive Summary

This diagnostic synthesizes **10,000** municipal service requests within the selected filter scope. Of these records, **9,778** have been resolved, and **222** remain active or in-progress. The historical median resolution duration across eligible closed cases is **8.7 hours** (0.4 days). A total of **2,231** cases (22.8%) exceeded the project-defined long-duration threshold of **124.925 hours** (~5.2 days).

Total Intake Records	10,000	**Median Duration (Eligible)**	8.7h (0.4d)
Closed (Valid Timestamp)	9,778	**Unresolved / Unclosed Backlog**	222 (active + null date)
Service-Eligible Closed	9,777	**Exceeding 124.9h Benchmark**	2,231 (22.8%)

2. Top Complaint Categories & Borough Distribution

Historical demand patterns demonstrate pronounced heterogeneity by complaint category and municipal jurisdiction. Top complaint types represent the primary service volume drivers:

Complaint Category	Volume	Share of Filter	Median Duration
Illegal Parking	1,466	14.7%	1.3h
Noise - Residential	924	9.2%	0.8h
HEAT/HOT WATER	734	7.3%	37.8h
Blocked Driveway	508	5.1%	1.6h
Noise - Street/Sidewalk	421	4.2%	0.6h

Borough	Complaints	Median Duration	Delay Rate (>124.9h)
BROOKLYN	3,037	8.5h	23.8%
QUEENS	2,370	4.9h	16.0%
MANHATTAN	2,118	13.3h	28.3%
BRONX	2,084	14.7h	24.2%
STATEN ISLAND	383	15.3h	19.1%

3. Leakage-Free Machine Learning Performance

All models were trained exclusively on 2023 intake data ($N=4,711$) using verified intake-time features only. Hyperparameters and preferred pipelines were chosen strictly on 2024-H1 validation data ($N=2,437$). The metrics below reflect the **unbiased final test evaluation** on held-out 2024-H2 records ($N=2,635$):

Task & Pipeline	Primary Metric	Secondary Metrics	Operational Assessment
Classification Logistic Regression (Class-Balanced)	**F1: 0.7093** Recall: 88.7%	Precision: 59.1% ROC-AUC: 0.9262 PR-AUC: 0.7227	Captured 488 of 550 delayed requests; generates 338 false alarms. Suited for screening and review triage.
Regression Random Forest (Log1p Target)	**MAE: 251.99h** MedAE: 15.59h	RMSE: 947.67h R^2: 0.1446	Typical absolute error is ~15.6 hours; heavy right-skew extreme outliers expand mean absolute error.

4. K-Means Operational Pattern Archetypes

K-Means clustering ($k=5$, Silhouette score: **0.3836**) grouped 274 municipal complaint segments into distinct operational demand archetypes:

Cluster Archetype	Segments	Avg Median Hours	Delay Rate %	Representative Categories
Cluster 0: High-Volume Rapid Municipal Dispatch	14 (5.1%)	9.4h (0.4d)	0.2%	Illegal Parking, Noise - Residential, HEAT/HOT WATER
Cluster 1: Moderate-Volume Routine Maintenance	129 (47.1%)	22.2h (0.9d)	7.5%	Abandoned Vehicle, Dirty Condition, Noise - Commercial
Cluster 2: Multi-Month Capital & Forestry Programs	8 (2.9%)	8903.9h (371.0d)	100.0%	New Tree Request, Food Establishment, Mobile Food Vendor
Cluster 3: Backlog-Prone / Specialized Operations	16 (5.8%)	517.3h (21.6d)	53.0%	Damaged Tree, Dead/Dying Tree, Noise - Helicopter
Cluster 4: Complex Residential Code Enforcement	107 (39.1%)	495.8h (20.7d)	62.0%	UNSANITARY CONDITION, APPLIANCE, DOOR/WINDOW

5. Scientific Limitations & Governance Disclaimers

1. Statistical Dependence, Not Causation: High feature importance scores indicate mathematical correlation within historical ticket routing, not physical root causes of delay.
2. Project-Defined Benchmark vs. Official SLA: The 124.925-hour cutoff is an empirical 75th percentile from the 2023 training partition, not a regulatory standard or official agency SLA violation.
3. Geographic & Operational Specificity: Models are trained on NYC 311 municipal data. They are not calibrated, validated, or representative for Indian municipal corporations (e.g., PCMC).
4. Screening Tool Notice: The classifier exhibits high recall (88.7%) with moderate precision (59.1%). It should be deployed solely for supervisor screening and review prioritization, not automated punitive actions.

6. Appendix: Reinforcement Learning & Queue Allocation Simulation

Simulation Parameters: Horizon: **10 periods** | Capacity: **5 slots/period** | Arrival Batch: **6 items/period** | Training: **15 episodes** (α=0.1, γ=0.95).
The multi-queue simulator evaluated 5 policies across identical complaint arrival streams over 5 random seeds:

Policy	Mean Reward	Slot Utilization	Items Handled	Mean Wait	Max Starve	True Long Capture
Q-Learning	37.33 ± 2.69	100.0%	50	0.78p	2.4p	**82.2%**
FIFO	38.31 ± 4.08	100.0%	50	0.82p	1.8p	**83.4%**
Risk-First	38.31 ± 4.82	100.0%	50	0.47p	6.6p	**98.8%**
Risk-Plus-Wait	39.93 ± 4.61	100.0%	50	0.81p	2.4p	**86.0%**
Round-Robin	31.55 ± 3.85	100.0%	50	0.67p	1.0p	**72.6%**

Disclaimer: Simulated waiting periods and retrospective outcome captures demonstrate policy focus under synthetic resource constraints. They do not prove that municipal resolution times would have been causally reduced in live operations.